

Task 1: Connect to a Public API

- **Instructions:**
 - Use Python (or Node.js/JavaScript) to connect to a public API (e.g., OpenWeatherMap or JSONPlaceholder).
 - Send a simple GET request to retrieve data.
 - Display the response in a readable format (pretty JSON).
- **Expected Output:**
 - Program prints API response (e.g., weather data or posts from JSONPlaceholder).
 - Output should be properly formatted JSON (not raw).

Prompt: write a python function that display weather details of a city using weather api without error handling. Display weather details as JSON output

To use this code:

1. First, you'll need to sign up for a free API key at Open Weather Map: <https://openweathermap.org/api>
2. Replace 'YOUR_API_KEY' in the code with your actual API key
3. Install the required requests package by running:

pip install requests

You can then run the script and enter any city name to get its current weather details. The output will look something like this

```
import requests
import json

def get_weather(city_name, api_key):
    base_url = "https://api.openweathermap.org/data/2.5/weather"
    complete_url = f"{base_url}?q={city_name}&appid={api_key}&uni
    response = requests.get(complete_url)
    weather_data = response.json()

    # Save the JSON output to a file
    with open("weather_output.json", "w") as json_file:
        json.dump(weather_data, json_file, indent=4)

    # Display formatted JSON output
    print(json.dumps(weather_data, indent=4))

if __name__ == "__main__":
    api_key = input("Enter your OpenWeatherMap API key: ")
    city = input("Enter city name: ")
    get_weather(city, api_key)
```

OUTPUT:

4/task1.py

Enter your OpenWeatherMap API key: 26cf5949c9080ab51db06364f54cc559

Enter city name: london

```
{
  "coord": {
    "lon": -0.1257,
    "lat": 51.5085
  },
  "weather": [
    {
      "id": 501,
      "main": "Rain",
      "description": "moderate rain",
      "icon": "10d"
    }
  ],
  "base": "stations",
  "main": {
    "temp": 12.11,
    "feels_like": 11.75,
    "temp_min": 11.42,
    "temp_max": 12.92,
    "pressure": 999,
    "humidity": 91,
    "sea_level": 999,
    "grnd_level": 995
  },
  "visibility": 8000,
  "wind": {
    "speed": 1.54,
    "deg": 200
  },
  "rain": {
    "1h": 2.73
  },
  "clouds": {
```

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Task 2: Add Error Handling for Invalid API Calls

- **Instructions:**
 - **Modify your code from Task 1 to handle errors.**
 - **Include try/except (Python) or try/catch (JavaScript) blocks.**
 - **Handle cases like:**
 - **Invalid URL**
 - **Network timeout**
 - **Wrong API key (if required)**
 - **Print user-friendly error messages.**
- **Expected Output:**
 - **If the API works, the result is shown as in Task 1.**
 - **If there's an error, output:**

Error: Could not connect to API. Check your API key or network connection.

Prompt: write a python function that display weather details of a city using weather api with error handling. Display weather details as JSON output

To use this code:

1. **First, you'll need to sign up for a free API key at Open Weather Map: <https://openweathermap.org/api>**
2. **Replace 'YOUR_API_KEY' in the code with your actual API key**
3. **Install the required requests package by running:**

```

import requests
import json

def get_weather(city_name, api_key):
    base_url = "https://api.openweathermap.org/data/2.5/weather"
    complete_url = f"{base_url}?q={city_name}&appid={api_key}&uni

    try:
        # Try to connect to the API
        response = requests.get(complete_url, timeout=5)
        response.raise_for_status() # Raise error for bad respon

        # Convert response to JSON
        weather_data = response.json()

        # Display formatted JSON output in terminal
        print(json.dumps(weather_data, indent=4))

        # ----- Create and save JSON file -----
        file_name = f"weather_{city_name}.json"
        with open(file_name, "w") as file:
            json.dump(weather_data, file, indent=4)

        print(f"\n✅ Weather data saved to '{file_name}'")

    except requests.exceptions.HTTPError:
        print("Error: Invalid response from the API. Please check
    except requests.exceptions.ConnectionError:
        print("Error: Could not connect to API. Check your intern
    except requests.exceptions.Timeout:
        print("Error: The request timed out. Try again later.")
    except requests.exceptions.RequestException:
        print("Error: Could not connect to API. Check your API ke

```

OUTPUT:

```
PS C:\Users\Admin\OneDrive\Desktop\AIAC\Lab-18.4> & C:/Users/Admin/AppData/Local/Programs/Python/Python313/python.exe c:/Users/Admin/OneDrive/Desktop/AIAC/Lab-18.4/Task2.py
```

```
{
  "base": "stations",
  "main": {
    "temp": 27.44,
    "feels_like": 27.85,
    "temp_min": 27.44,
    "temp_max": 27.44,
    "pressure": 1009,
    "humidity": 50,
    "sea_level": 1009,
    "grnd_level": 990
  },
  "visibility": 10000,
  "wind": {
    "speed": 2.24,
    "deg": 86,
    "gust": 2.86
  },
  "clouds": {
    "all": 96
  },
  "dt": 1761740321,
  "sys": {
    "country": "IN",
    "sunrise": 1761699353,
    "sunset": 1761739634
  },
  "timezone": 19800,
  "id": 1279259,
  "name": "Agra",
  "cod": 200
}
```

```
✓ Weather data saved to 'weather_agra.json'
```

```
PS C:\Users\Admin\OneDrive\Desktop\AIAC\Lab-18.4>
```

Task 3: Extract and Display Specific Data

- **Instructions:**
 1. From the API response (e.g., weather API), extract specific fields (temperature, humidity, description).
 2. Display them in a user-friendly format (not raw JSON).
- **Expected Output:**
- **City: London**
- **Temperature: 18°C**
- **Humidity: 60%**
- **Weather: clear sky**

```

import requests
import json

def get_weather(city_name, api_key):
    base_url = "https://api.openweathermap.org/data/2.5/weather"
    complete_url = f"{base_url}?q={city_name}&appid={api_key}&uni

    try:
        # Connect to the API
        response = requests.get(complete_url, timeout=5)
        response.raise_for_status()

        # Convert response to JSON
        weather_data = response.json()

        # ----- Extract specific fields -----
        city = weather_data["name"]
        temperature = weather_data["main"]["temp"]
        humidity = weather_data["main"]["humidity"]
        description = weather_data["weather"][0]["description"]

        # ----- Display in user-friendly format -----
        print(f"\nCity: {city}")
        print(f"Temperature: {temperature}°C")
        print(f"Humidity: {humidity}%")
        print(f"Weather: {description.capitalize()}")

        # Save entire response as JSON (optional)
        with open(f"weather_{city}.json", "w") as file:
            json.dump(weather_data, file, indent=4)

    except requests.exceptions.HTTPError:
        print("Error: Invalid response from the API. Please check
    except requests.exceptions.ConnectionError:

```


OUTPUT:

```
4/Task3.py
Enter your OpenWeatherMap API key: 26cf5949c9080ab51db06364f54cc559
Enter city name: london

City: London
Temperature: 12.14°C
Humidity: 91%
Weather: Moderate rain
PS C:\Users\Admin\OneDrive\Desktop\AIAC\Lab-18.4>
```

Task 4: Build a Function with Parameters

- **Instructions:**
 - Write a function that accepts a parameter (e.g., city name for weather API).
 - The function should call the API dynamically based on user input.
 - Include error handling if the city is invalid.
- **Expected Output:**
 - Input: "New York"
 - **Output:**
 - City: New York
 - Temperature: 22°C
 - Humidity: 55%
 - Weather: Few clouds
 - Input: "xyz123"
 - **Output:**

Error: City not found. Please enter a valid city.

Prompt: write a python function that display weather details of a city using weather api with error handling. Display weather details in user friendly format

```

import requests

def get_weather(city_name, api_key):
    base_url = "https://api.openweathermap.org/data/2.5/weather"
    complete_url = f"{base_url}?q={city_name}&appid={api_key}&units={units}"

    try:
        response = requests.get(complete_url, timeout=5)
        response.raise_for_status() # Raises error for 4xx/5xx status codes

        data = response.json()

        # Check if the API returned a valid city
        if data.get("cod") != 200:
            print("Error: City not found. Please enter a valid city name.")
            return

        # Extract specific weather details
        city = data["name"]
        temp = data["main"]["temp"]
        humidity = data["main"]["humidity"]
        description = data["weather"][0]["description"]

        # Display in user-friendly format
        print(f"\nCity: {city}")
        print(f"Temperature: {temp}°C")
        print(f"Humidity: {humidity}%")
        print(f"Weather: {description.capitalize()}")

    except requests.exceptions.ConnectionError:
        print("Error: Could not connect to API. Check your internet connection.")
    except requests.exceptions.Timeout:
        print("Error: The request timed out. Try again later.")
    except requests.exceptions.RequestException:
        print("Error: An unexpected error occurred.")

```

OUTPUT:

```

PS C:\Users\Admin\OneDrive\Desktop\AIAC\Lab-18.4> & C:/Users/Admin/AppData/Local/
Programs/Python/Python313/python.exe c:/Users/Admin/OneDrive/Desktop/AIAC/Lab-18.
4/Task4.py
Enter your OpenWeatherMap API key: 26cf5949c9080ab51db06364f54cc559
Enter city name: new york

City: New York
Temperature: 10.12°C
Humidity: 70%
Weather: Overcast clouds
PS C:\Users\Admin\OneDrive\Desktop\AIAC\Lab-18.4>

```

Task 5: Store API Results Locally

- Instructions:
 - Extend your function from Task 4.
 - Save the extracted API results into a local file (results.json or results.txt).
 - Each new request should append results without overwriting old ones.
- Expected Output:
 - Console still shows formatted output.
 - A local file results.json is created/updated with stored responses like:
 - [
 - {"city": "London", "temp": 18, "humidity": 60, "weather": "Clear sky"},
 - {"city": "New York", "temp": 22, "humidity": 55, "weather": "Few clouds"}
 -]

Prompt: write a python function that display weather details of a city using weather api with error handling. Display weather details as JSON output. Store the weather details in current directory as text file, every run output will append

Deliverables (For All Tasks)

1. AI-generated prompts for code and test case generation.
2. At least 3 assert test cases for each task.
3. AI-generated initial code and execution screenshots.

4. Analysis of whether code passes all tests.
5. Improved final version with inline comments and explanation.
6. Compiled report (Word/PDF) with prompts, test cases, assertions, code, and output.

```
import requests
import json
import os

def get_weather(city_name, api_key):
    base_url = "https://api.openweathermap.org/data/2.5/weather"
    complete_url = f"{base_url}?q={city_name}&appid={api_key}&uni

    try:
        # Send API request
        response = requests.get(complete_url, timeout=5)
        response.raise_for_status()

        data = response.json()

        # Check if the city is valid
        if data.get("cod") != 200:
            print("Error: City not found. Please enter a valid ci
            return

        # Extract important details
        city = data["name"]
        temp = data["main"]["temp"]
        humidity = data["main"]["humidity"]
        description = data["weather"][0]["description"]

        # Display formatted result in console
        print(f"\nCity: {city}")
        print(f"Temperature: {temp}°C")
        print(f"Humidity: {humidity}%")
        print(f"Weather: {description.capitalize()}")

        # Prepare data to store
        new_entry = f
```

OUTPUT:

```
PS C:\Users\Admin\OneDrive\Desktop\AIAC\Lab-18.4> & C:/Users/Admin/AppData/Local/Programs/Python/Python313/python.exe c:/Users/Admin/OneDrive/Desktop/AIAC/Lab-18.4/Task5.py
Enter your OpenWeatherMap API key: 26cf5949c9080ab51db06364f54cc559
Enter city name: karimnagar

City: Karimnagar
Temperature: 23.36°C
Humidity: 97%
Weather: Heavy intensity rain

✅ Weather data for Karimnagar saved to 'results.json'
PS C:\Users\Admin\OneDrive\Desktop\AIAC\Lab-18.4>
```